

Newman-Penrose Formalism Calculations

Stephani et al., Equation (35.30_e0)

Metric

$$ds^2 = -2 \left(A - \frac{b_x^2 + b_y^2}{P} \right) du^2 - 2 du dv + \frac{4b_x}{P} du dx + \frac{4b_y}{P} du dy + \frac{2}{P} (dx^2 + dy^2).$$

Coordinates

$$(x^\mu) = (u, v, x, y)$$

Metric Functions

$$K = K(u),$$

$$H = H(u, x, y),$$

$$Q = 1,$$

$$P = k^2 Q^2,$$

$$A = lv + H,$$

$$b_x = b_x(u, x, y),$$

$$b_y = b_y(u, x, y).$$

Null Tetrad

Covariant components

$$l = (1) du$$

$$n = \left(lv - \frac{b_x(u, x, y)^2}{k^2} - \frac{b_y(u, x, y)^2}{k^2} + H \right) du + (1) dv + \left(-\frac{2b_x(u, x, y)}{k^2} \right) dx + \left(-\frac{2b_y(u, x, y)}{k^2} \right) dy$$

$$m = \left(\sqrt{\frac{1}{k^2}} \right) dx + \left(i \sqrt{\frac{1}{k^2}} \right) dy$$

$$\bar{m} = \left(\sqrt{\frac{1}{k^2}} \right) dx + \left(-i \sqrt{\frac{1}{k^2}} \right) dy$$

Contravariant components

$$l = (-1) dv$$

$$n = (-1) du + \left(lv - \frac{b_x(u, x, y)^2}{k^2} - \frac{b_y(u, x, y)^2}{k^2} + H \right) dv$$

$$m = \left(\frac{b_x(u, x, y)}{|k|} + \frac{i b_y(u, x, y)}{|k|} \right) dv + \left(\frac{1}{2} |k| \right) dx + \left(\frac{1}{2} i |k| \right) dy$$

$$\bar{m} = \left(\frac{b_x(u, x, y)}{|k|} - \frac{i b_y(u, x, y)}{|k|} \right) dv + \left(\frac{1}{2} |k| \right) dx + \left(-\frac{1}{2} i |k| \right) dy$$

Directional Derivatives

$$D = -\partial_v$$

$$\Delta = -\partial_u + lv - \frac{b_x(u, x, y)^2}{k^2} - \frac{b_y(u, x, y)^2}{k^2} + H \partial_v$$

$$\delta = \frac{b_x(u, x, y)}{|k|} + \frac{i b_y(u, x, y)}{|k|} \partial_v + \frac{1}{2} |k| \partial_x + \frac{1}{2} i |k| \partial_y$$

$$\bar{\delta} = \frac{b_x(u, x, y)}{|k|} - \frac{i b_y(u, x, y)}{|k|} \partial_v + \frac{1}{2} |k| \partial_x - \frac{1}{2} i |k| \partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{\frac{\partial}{\partial u} k}{k} - \frac{1}{2} i \frac{\partial}{\partial y} b_x(u, x, y) + \frac{1}{2} i \frac{\partial}{\partial x} b_y(u, x, y)$$

$$\lambda = 0$$

$$\begin{aligned} \nu = & \frac{k^2 \frac{\partial}{\partial x} H}{2 |k|} - \frac{i k^2 \frac{\partial}{\partial y} H}{2 |k|} + \frac{l b_x(u, x, y)}{|k|} - \frac{i l b_y(u, x, y)}{|k|} - \frac{b_x(u, x, y) \frac{\partial}{\partial x} b_x(u, x, y)}{|k|} + \frac{i b_x(u, x, y) \frac{\partial}{\partial y} b_x(u, x, y)}{|k|} - \frac{b_y(u, x, y) \frac{\partial}{\partial x} b_y(u, x, y)}{|k|} \\ & + \frac{i b_y(u, x, y) \frac{\partial}{\partial y} b_y(u, x, y)}{|k|} + \frac{\frac{\partial}{\partial u} b_x(u, x, y)}{|k|} - \frac{i \frac{\partial}{\partial u} b_y(u, x, y)}{|k|} - \frac{2 b_x(u, x, y) \frac{\partial}{\partial u} k}{|k| k} + \frac{2 i b_y(u, x, y) \frac{\partial}{\partial u} k}{|k| k} \end{aligned}$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = -\frac{1}{2} l - \frac{1}{4} i \frac{\partial}{\partial y} b_x(u, x, y) + \frac{1}{4} i \frac{\partial}{\partial x} b_y(u, x, y)$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = -\frac{1}{8}i|k|\frac{\partial^2}{\partial x\partial y}b_x(u, x, y) + \frac{1}{8}|k|\frac{\partial^2}{(\partial y)^2}b_x(u, x, y) + \frac{1}{8}i|k|\frac{\partial^2}{(\partial x)^2}b_y(u, x, y) - \frac{1}{8}|k|\frac{\partial^2}{\partial x\partial y}b_y(u, x, y)$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = \frac{1}{8}i|k|\frac{\partial^2}{\partial x\partial y}b_x(u, x, y) + \frac{1}{8}|k|\frac{\partial^2}{(\partial y)^2}b_x(u, x, y) - \frac{1}{8}i|k|\frac{\partial^2}{(\partial x)^2}b_y(u, x, y) - \frac{1}{8}|k|\frac{\partial^2}{\partial x\partial y}b_y(u, x, y)$$

$$\begin{aligned}\Phi_{22} = & \frac{1}{4}k^2\frac{\partial^2}{(\partial x)^2}H + \frac{1}{4}k^2\frac{\partial^2}{(\partial y)^2}H + \frac{1}{2}l\frac{\partial}{\partial x}b_x(u, x, y) - \frac{1}{2}\frac{\partial}{\partial x}b_x(u, x, y)^2 - \frac{1}{2}b_x(u, x, y)\frac{\partial^2}{(\partial x)^2}b_x(u, x, y) - \frac{1}{4}\frac{\partial}{\partial y}b_x(u, x, y)^2 - \frac{1}{2}b_x(u, x, y)\frac{\partial^2}{(\partial y)^2}b_x(u, x, y) \\ & - \frac{1}{2}\frac{\partial}{\partial y}b_x(u, x, y)\frac{\partial}{\partial x}b_y(u, x, y) - \frac{1}{4}\frac{\partial}{\partial x}b_y(u, x, y)^2 - \frac{1}{2}b_y(u, x, y)\frac{\partial^2}{(\partial x)^2}b_y(u, x, y) + \frac{1}{2}l\frac{\partial}{\partial y}b_y(u, x, y) - \frac{1}{2}\frac{\partial}{\partial y}b_y(u, x, y)^2 \\ & - \frac{1}{2}b_y(u, x, y)\frac{\partial^2}{(\partial y)^2}b_y(u, x, y) + \frac{l\frac{\partial}{\partial u}k}{k} - \frac{\frac{\partial}{\partial x}b_x(u, x, y)\frac{\partial}{\partial u}k}{k} - \frac{\frac{\partial}{\partial y}b_y(u, x, y)\frac{\partial}{\partial u}k}{k} - \frac{2\frac{\partial}{\partial u}k^2}{k^2} + \frac{\frac{\partial^2}{(\partial u)^2}k}{k} + \frac{1}{2}\frac{\partial^2}{\partial u\partial x}b_x(u, x, y) + \frac{1}{2}\frac{\partial^2}{\partial u\partial y}b_y(u, x, y)\end{aligned}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = 0$$

$$\Psi_3 = -\frac{1}{8}i|k|\frac{\partial^2}{\partial x\partial y}b_x(u, x, y) - \frac{1}{8}|k|\frac{\partial^2}{(\partial y)^2}b_x(u, x, y) + \frac{1}{8}i|k|\frac{\partial^2}{(\partial x)^2}b_y(u, x, y) + \frac{1}{8}|k|\frac{\partial^2}{\partial x\partial y}b_y(u, x, y)$$

$$\begin{aligned}\Psi_4 = & \frac{1}{4}k^2\frac{\partial^2}{(\partial x)^2}H - \frac{1}{2}ik^2\frac{\partial^2}{\partial x\partial y}H - \frac{1}{4}k^2\frac{\partial^2}{(\partial y)^2}H + \frac{1}{2}l\frac{\partial}{\partial x}b_x(u, x, y) - \frac{1}{2}\frac{\partial}{\partial x}b_x(u, x, y)^2 - \frac{1}{2}b_x(u, x, y)\frac{\partial^2}{(\partial x)^2}b_x(u, x, y) + ib_x(u, x, y)\frac{\partial^2}{\partial x\partial y}b_x(u, x, y) - \frac{1}{2}il\frac{\partial}{\partial y}b_x(u, x, y) \\ & + i\frac{\partial}{\partial x}b_x(u, x, y)\frac{\partial}{\partial y}b_x(u, x, y) + \frac{1}{2}\frac{\partial}{\partial y}b_x(u, x, y)^2 + \frac{1}{2}b_x(u, x, y)\frac{\partial^2}{(\partial y)^2}b_x(u, x, y) - \frac{1}{2}il\frac{\partial}{\partial x}b_y(u, x, y) - \frac{1}{2}\frac{\partial}{\partial x}b_y(u, x, y)^2 - \frac{1}{2}b_y(u, x, y)\frac{\partial^2}{(\partial x)^2}b_y(u, x, y) \\ & + ib_y(u, x, y)\frac{\partial^2}{\partial x\partial y}b_y(u, x, y) - \frac{1}{2}l\frac{\partial}{\partial y}b_y(u, x, y) + i\frac{\partial}{\partial x}b_y(u, x, y)\frac{\partial}{\partial y}b_y(u, x, y) + \frac{1}{2}\frac{\partial}{\partial y}b_y(u, x, y)^2 + \frac{1}{2}b_y(u, x, y)\frac{\partial^2}{(\partial y)^2}b_y(u, x, y) - \frac{\frac{\partial}{\partial x}b_x(u, x, y)\frac{\partial}{\partial u}k}{k} \\ & + \frac{i\frac{\partial}{\partial y}b_x(u, x, y)\frac{\partial}{\partial u}k}{k} + \frac{i\frac{\partial}{\partial x}b_y(u, x, y)\frac{\partial}{\partial u}k}{k} + \frac{\frac{\partial}{\partial y}b_y(u, x, y)\frac{\partial}{\partial u}k}{k} + \frac{1}{2}\frac{\partial^2}{\partial u\partial x}b_x(u, x, y) - \frac{1}{2}i\frac{\partial^2}{\partial u\partial y}b_x(u, x, y) - \frac{1}{2}i\frac{\partial^2}{\partial u\partial x}b_y(u, x, y) - \frac{1}{2}\frac{\partial^2}{\partial u\partial y}b_y(u, x, y)\end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

$$\begin{aligned}BI4 = & -\frac{i\frac{\partial}{\partial u}k\frac{\partial^2}{\partial x\partial y}b_x(u, x, y)}{8|k|} - \frac{\frac{\partial}{\partial u}k\frac{\partial^2}{(\partial y)^2}b_x(u, x, y)}{8|k|} + \frac{i\frac{\partial}{\partial u}k\frac{\partial^2}{(\partial x)^2}b_y(u, x, y)}{8|k|} + \frac{\frac{\partial}{\partial u}k\frac{\partial^2}{\partial x\partial y}b_y(u, x, y)}{8|k|} - \frac{i\bar{k}\frac{\partial^2}{\partial x\partial y}b_x(u, x, y)\frac{\partial}{\partial u}k}{8|k|} + \frac{ik\frac{\partial^2}{\partial x\partial y}b_x(u, x, y)\frac{\partial}{\partial u}k}{4|k|} \\ & - \frac{\bar{k}\frac{\partial^2}{(\partial y)^2}b_x(u, x, y)\frac{\partial}{\partial u}k}{8|k|} + \frac{k\frac{\partial^2}{(\partial y)^2}b_x(u, x, y)\frac{\partial}{\partial u}k}{4|k|} + \frac{i\bar{k}\frac{\partial^2}{(\partial x)^2}b_y(u, x, y)\frac{\partial}{\partial u}k}{8|k|} - \frac{ik\frac{\partial^2}{(\partial x)^2}b_y(u, x, y)\frac{\partial}{\partial u}k}{4|k|} + \frac{\bar{k}\frac{\partial^2}{\partial x\partial y}b_y(u, x, y)\frac{\partial}{\partial u}k}{8|k|} - \frac{k\frac{\partial^2}{\partial x\partial y}b_y(u, x, y)\frac{\partial}{\partial u}k}{4|k|}\end{aligned}$$

Equations are vanished.

Petrov Type

Type "III"